INDUSTRIAL ASSIGNMENT- 6 ER Diagram for the DB Design
STUDENT INTERNSHIP AND PLACEMENT MANAGEMENT PORTAL

1. Introduction

The Student Internship and Placement Management Portal is a centralized digital platform that supports the complete internship and campus placement cycle of a diploma college. The supplied project documents describe user accounts and access, student profiles and resumes, companies and opportunities, applications, interviews and offers, dashboards, reports, and search and filter facilities. These activities depend on a consistent structure for storing and linking records.

An Entity-Relationship (ER) model provides a structured view of the database entities, their attributes, key fields, and relationships. It helps make data ownership and references visible before implementation, and allows the SQL design created in Industrial Assignment 5 to be reviewed without changing the project scope.

2. Objective of the Assessment

The objective of this assessment is to document the database schema represented by the SQL definitions of Industrial Assignment 5 and to cross-check the entity and attribute inventory against the Entity and Attribute Dictionary given there. The diagrams identify primary keys, explicitly declared foreign keys, relevant uniqueness constraints, and relationship cardinalities. They are a representation of the supplied design, not a proposal to redesign it.

3. Scope of the Database Design

The assessment covers six database modules: EP01 User Management; EP02 Student Profile Management; EP03 Company and Opportunity Management; EP04 Application Management; EP05 Interview and Offer Management; and EP06 Dashboard and Reports. EP07 Search and Filter is represented through the existing opportunities, student profiles and applications records and does not add a separate table. The schema includes 16 unique entities.

The relationship diagrams draw solid connectors only where the SQL CREATE TABLE statements explicitly declare a FOREIGN KEY. The schema declares 28 foreign-key constraints and two composite UNIQUE constraints: UNIQUE(student_id, opportunity_id) in applications and UNIQUE(user_id, widget_name) in dashboard_preferences. Users and opportunities are removed by changing their status to “Deleted” (soft delete), so history and application records always keep a valid parent.

4. Database Entity Inventory (1 of 2)

The following inventory uses the unique tables created in Industrial Assignment 5. Primary key names are copied from the corresponding SQL definitions. Descriptions summarize each table’s documented purpose.

S. No.	Entity / Table	Episode / Module	Primary Key	Purpose
1	users	EP01 - User Management	user_id	Stores user identity, contact details, role, status, login-lock data and role-specific registration information.
2	login_sessions	EP01 - User Management	session_id	Records login sessions, tokens, login/logout times and last activity.
3	user_access_history	EP01 - User Management	history_id	Stores approval, status-change, role-change and deletion audit details for accounts.
4	student_profiles	EP02 - Student Profile Management	student_id	Stores the academic, skill and contact profile of a student and its verification status.
5	student_resumes	EP02 - Student Profile Management	resume_id	Stores resume file details of a student; one resume is marked current.
6	student_edit_history	EP02 - Student Profile Management	history_id	Records changes to individual student profile fields with old and new values.
7	companies	EP03 - Company and Opportunity Management	company_id	Stores the profile of a recruiting company linked to its approved recruiter account.
8	opportunities	EP03 - Company and Opportunity Management	opportunity_id	Stores internship and placement openings with eligibility, compensation, deadline and status.

4. Database Entity Inventory (2 of 2)

S. No.	Entity / Table	Episode / Module	Primary Key	Purpose

S. No.	Entity / Table	Episode / Module	Primary Key	Purpose
9	opportunity_edit_history	EP03 - Company and Opportunity Management	history_id	Records changes to opportunity fields with old and new values and the editor.
10	opportunity_deletion_history	EP03 - Company and Opportunity Management	deletion_id	Stores opportunity deletion and restoration details with a copy of the title.
11	applications	EP04 - Application Management	application_id	Stores a student's application to an opportunity and its current status; links students and opportunities.
12	application_status_history	EP04 - Application Management	history_id	Tracks every application status change and the user who made it.
13	interviews	EP05 - Interview and Offer Management	interview_id	Stores interview schedule, mode, venue or link, round and result for an application.
14	offers	EP05 - Interview and Offer Management	offer_id	Stores the offer released for a selected application and the student's response.
15	dashboard_preferences	EP06 - Dashboard and Reports	preference_id	Stores per-user dashboard widget visibility and display ordering.
16	report_history	EP06 - Dashboard and Reports	report_id	Records report type, date range, scope, export format and generation details.

5. ER Diagram Notation and Symbols

Notation	Meaning in this assessment
Entity box	A rectangle represents one table; its grey heading is the exact table name and its rows list attributes with MySQL data types. In the overall diagram, boxes are coloured by module and show only the primary key.
PK	Primary key declared in the SQL. PKs identify rows and are not nullable.
FK	Foreign key explicitly declared with FOREIGN KEY (...) REFERENCES ... in the SQL. Columns without a declaration are not drawn as relations.
NN / UQ / AI	NN = NOT NULL; UQ = UNIQUE; AI = AUTO_INCREMENT. Unmarked non-key attributes may accept NULL unless their SQL definition says otherwise.
|| (double bar)	Exactly one / mandatory participation at that end.
O| (circle and bar)	Zero or one / optional single participation.
O< (circle and crow's foot)	Zero or many / optional many participation.
Dashed box	A table that belongs to another module, shown for reference only so that a foreign key can be drawn.
Associative entity	A table that links two entity types. applications links student_profiles and opportunities; the SQL also declares UNIQUE(student_id, opportunity_id).

6. Overall Database ER Diagram

All 16 unique tables are shown. Connectors are drawn only for the 28 foreign keys explicitly declared in the SQL of Industrial Assignment 5. Colours indicate modules: green EP01, blue EP02, yellow EP03, orange EP04, purple EP05 and grey EP06.

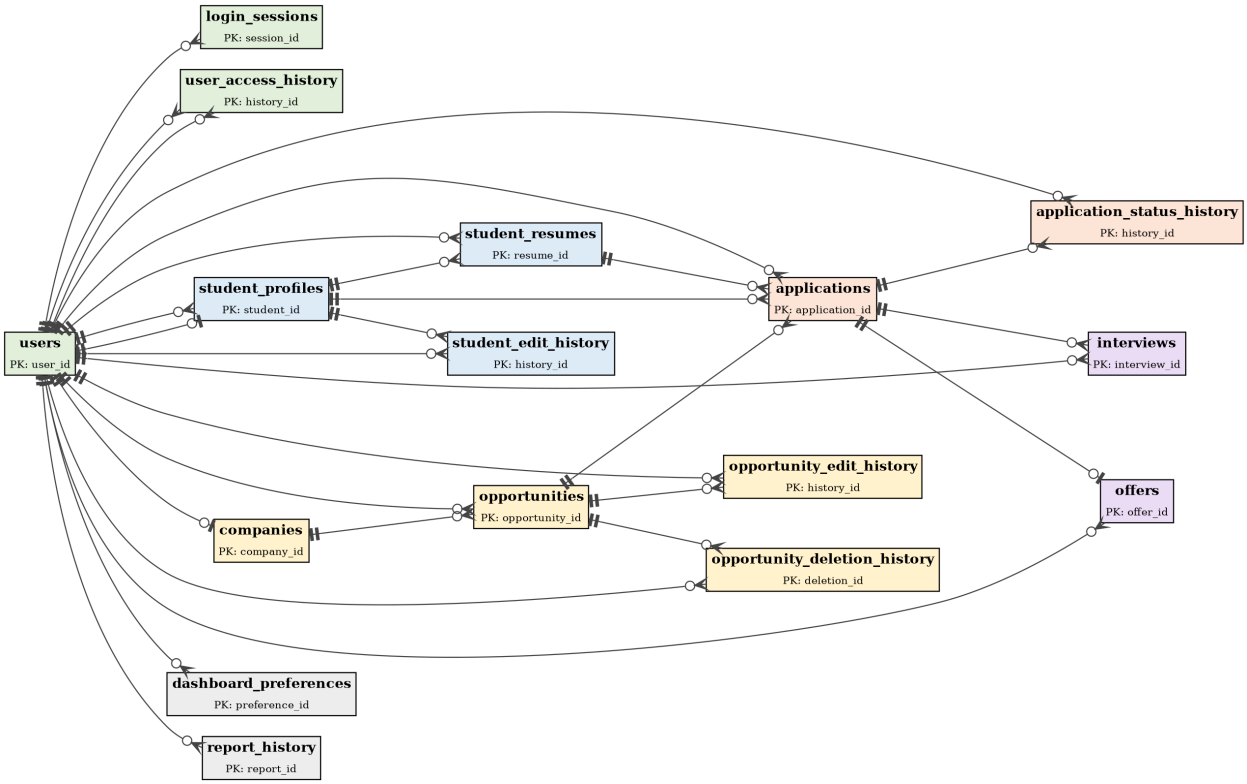


Figure 1: Overall Entity-Relationship Diagram of the Student Internship and Placement Management Portal

Crow's Foot ends: || = one; O| = zero-or-one; O< = zero-or-many. Because users is referenced by many tables (accounts, actors who performed an action and owners of a profile), many connectors meet at the users entity.

7.1 EP01 - User Management

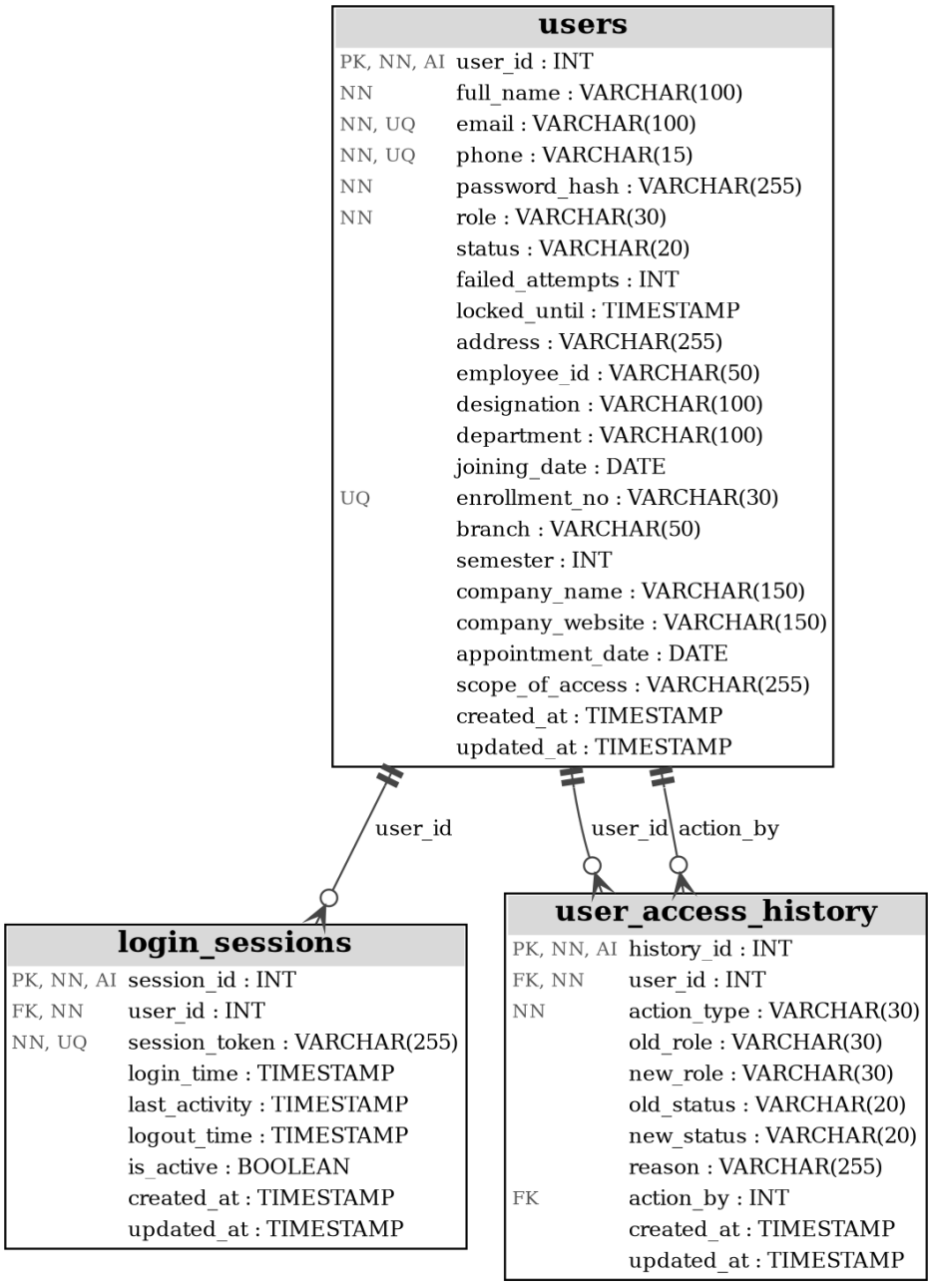


Figure 2: User Management ER Diagram

PK = primary key; FK = explicitly declared foreign key; NN = NOT NULL; UQ = UNIQUE; AI = AUTO_INCREMENT. Crow's Foot: || one; O| zero-or-one; O< zero-or-many. Unmarked non-key columns may accept NULL.

The users table stores role-specific registration fields (student, recruiter, staff and administrator) in one table, together with the status and login-lock columns. login_sessions references users

through user_id. user_access_history references users twice: user_id identifies the account that changed and action_by identifies the user who performed the change.

Only users.user_id is referenced by other modules; every other module in this document therefore connects back to this table.

7.2 EP02 - Student Profile Management

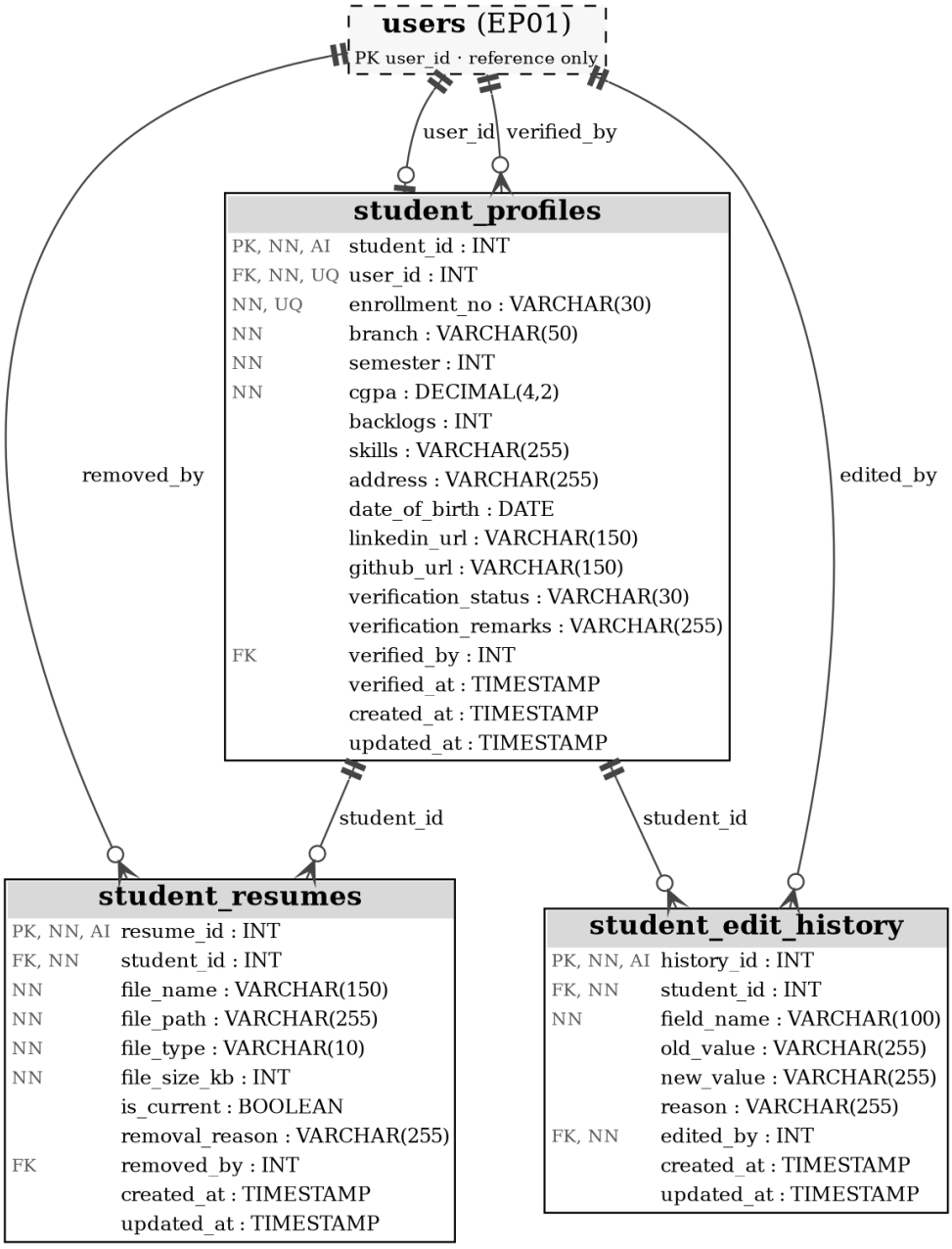


Figure 3: Student Profile Management ER Diagram

PK = primary key; FK = explicitly declared foreign key; NN = NOT NULL; UQ = UNIQUE; AI = AUTO_INCREMENT. Crow's Foot: || one; O| zero-or-one; O< zero-or-many. Unmarked non-key columns may accept NULL.

student_profiles has a one-to-one link with users through the UNIQUE foreign key user_id and carries the academic data and verification status. student_resumes and student_edit_history

reference student_profiles through student_id; removed_by, edited_by and verified_by reference the users who performed each action.

student_profiles.enrollment_no, branch and semester are copied from users when the profile is created, so the profile remains the source of truth for eligibility checks.

7.3 EP03 - Company and Opportunity Management

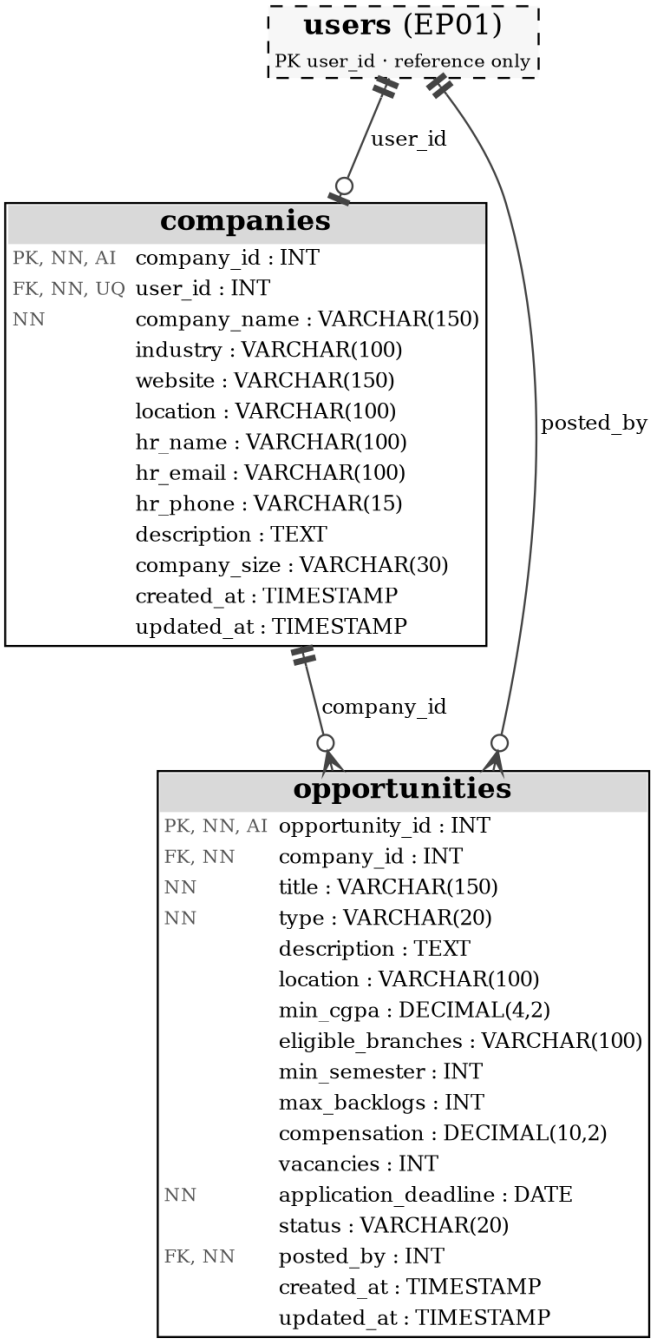


Figure 4a: Company and Opportunity ER Diagram – companies and opportunities

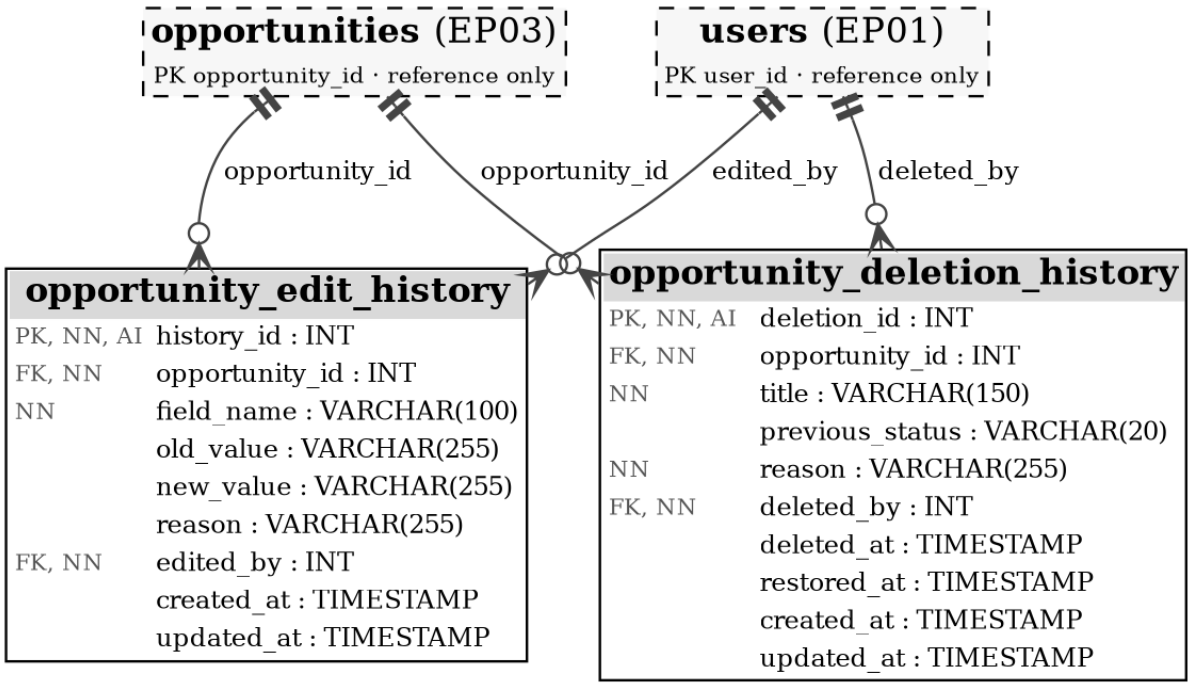


Figure 4b: Company and Opportunity ER Diagram – opportunity history tables

PK = primary key; FK = explicitly declared foreign key; NN = NOT NULL; UQ = UNIQUE; AI = AUTO_INCREMENT. Crow's Foot: || one; O| zero-or-one; O< zero-or-many. Unmarked non-key columns may accept NULL.

companies has a one-to-one link with the recruiter account through the UNIQUE foreign key user_id. opportunities references companies through company_id and records the poster in posted_by (a recruiter, or a Placement Officer posting on behalf of the company). opportunity_edit_history and opportunity_deletion_history reference opportunities and the users who made each change.

opportunity_deletion_history.previous_status stores the status before deletion so that a restored opportunity returns to its earlier state.

7.4 EP04 - Application Management

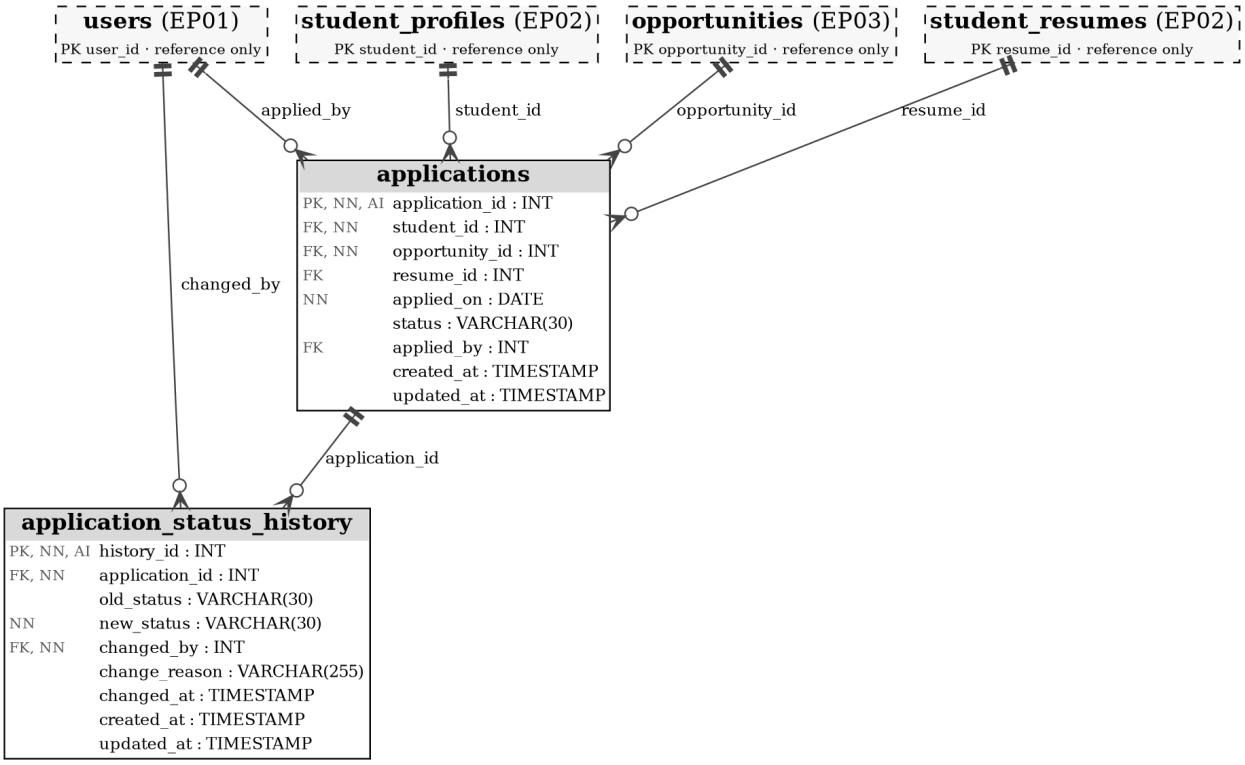


Figure 5: Application Management ER Diagram

PK = primary key; FK = explicitly declared foreign key; NN = NOT NULL; UQ = UNIQUE; AI = AUTO_INCREMENT. Crow's Foot: || one; O| zero-or-one; O< zero-or-many. Unmarked non-key columns may accept NULL.

applications is the associative entity between student_profiles and opportunities. Its UNIQUE(student_id, opportunity_id) constraint prevents duplicate applications. resume_id optionally references the resume attached to the application and applied_by optionally references the Placement Officer who applied on behalf of the student. application_status_history keeps every status change with the user who made it.

The status values (Applied, Shortlisted, Interview Scheduled, Selected, Rejected and Withdrawn) are stored as text and checked by the application rules of Industrial Assignments 2–4.

7.5 EP05 - Interview and Offer Management

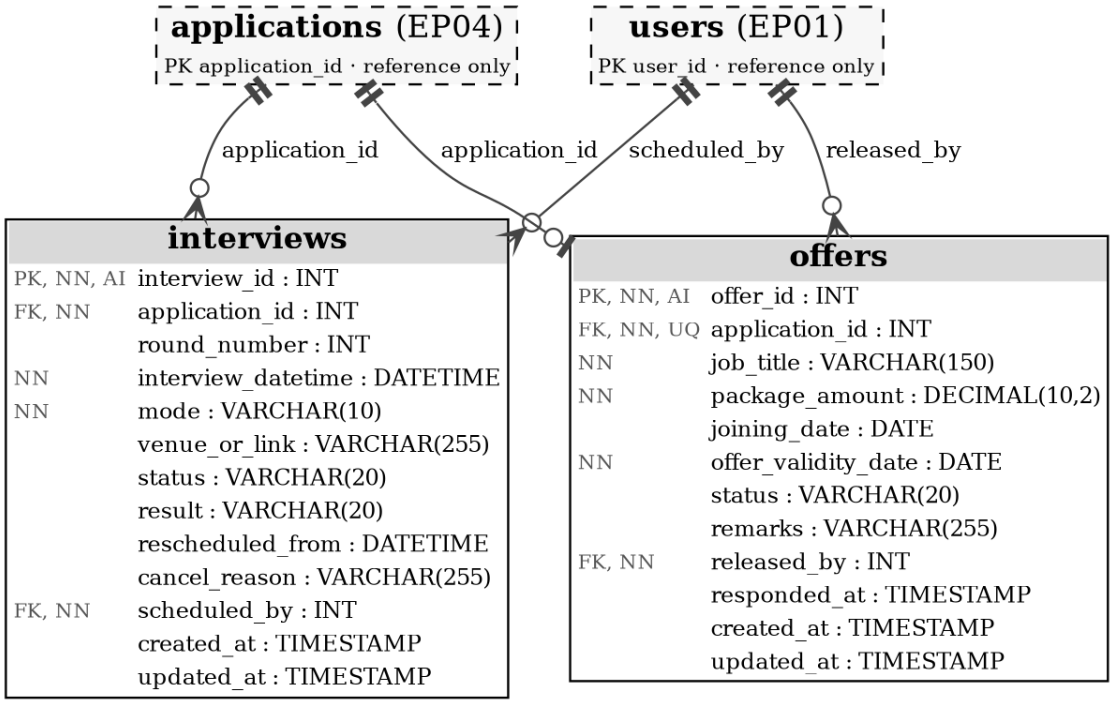


Figure 6: Interview and Offer Management ER Diagram

PK = primary key; FK = explicitly declared foreign key; NN = NOT NULL; UQ = UNIQUE; AI = AUTO_INCREMENT. Crow's Foot: || one; O| zero-or-one; O< zero-or-many. Unmarked non-key columns may accept NULL.

interviews references applications through application_id, so one application can have several interview rounds. offers references applications through a UNIQUE application_id, so an application can have at most one offer. scheduled_by and released_by reference the users who scheduled the interview or released the offer.

An offer can be released only for an application whose status is Selected; this is enforced by the query logic of Industrial Assignment 5 and not by a database constraint.

7.6 EP06 - Dashboard and Reports

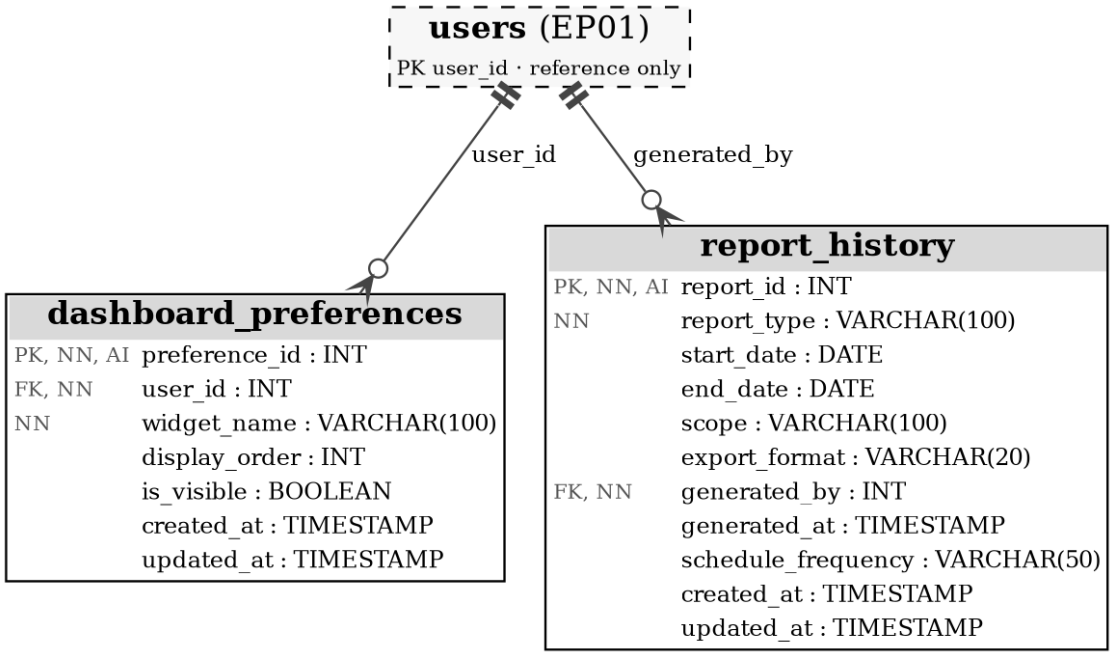


Figure 7: Dashboard and Reports ER Diagram

PK = primary key; FK = explicitly declared foreign key; NN = NOT NULL; UQ = UNIQUE; AI = AUTO_INCREMENT. Crow's Foot: || one; O| zero-or-one; O< zero-or-many. Unmarked non-key columns may accept NULL.

dashboard_preferences references users through user_id and stores one row per widget for each user (UNIQUE(user_id, widget_name)). report_history references users through generated_by and stores the report type, period, scope and format of each generated report.

Dashboard figures and reports are calculated from the other modules' tables and are not stored.

7.7 EP07 - Search and Filter

EP07 Search and Filter uses the existing opportunities, student_profiles, users, applications and companies tables and introduces no additional table. The following columns are the ones searched or filtered by Tasks T25–T28.

Task	Table	Columns Searched or Filtered
T25	opportunities, companies	opportunities.title, description, location; companies.company_name
T26	opportunities	type, status, location, min_cgpa, eligible_branches, min_semester, max_backlogs, application_deadline
T27	student_profiles, users	users.full_name; student_profiles.enrollment_no, branch, semester, cgpa, verification_status
T28	applications, student_profiles, opportunities, companies	applications.status, applied_on; student_profiles.branch, cgpa; opportunities.opportunity_id; companies.company_name

No dedicated search or filter table is created; every search and filter works directly on the module tables described in Sections 7.1 to 7.6.

8. Relationship and Cardinality Table (1 of 2)

“Parent” is the referenced table; “child FK” is the column that holds the reference. The many side is zero-or-many unless a constraint narrows it. Required FK columns mean each child row must reference one parent; nullable FK columns allow the child row to have no parent at that relationship end.

No.	Parent Entity	Child Entity	Child FK	Cardinality	Meaning
1	users	login_sessions	user_id	1 to 0..*	Each session references one user; a user may have many sessions.
2	users	user_access_history	user_id	1 to 0..*	Each history row references the account that was changed.
3	users	user_access_history	action_by	1 to 0..*; child FK optional	Optional reference to the user who performed the change (nullable).
4	users	student_profiles	user_id	1 to 0..1	Each student profile belongs to exactly one student user account; the FK is UNIQUE.
5	users	student_profiles	verified_by	1 to 0..*; child FK optional	Optional reference to the Placement Officer who verified the profile (nullable).

No.	Parent Entity	Child Entity	Child FK	Cardinality	Meaning
6	student_profiles	student_resumes	student_id	1 to 0..*	Each resume row belongs to one student profile; a student may have many versions.
7	users	student_resumes	removed_by	1 to 0..*; child FK optional	Optional reference to the user who removed the resume (nullable).
8	student_profiles	student_edit_history	student_id	1 to 0..*	Each edit row references one student profile.
9	users	student_edit_history	edited_by	1 to 0..*	Each edit row references the user who made the change.
10	users	companies	user_id	1 to 0..1	Each company profile belongs to exactly one recruiter account; the FK is UNIQUE.
11	companies	opportunities	company_id	1 to 0..*	Each opportunity belongs to one company; a company may post many opportunities.
12	users	opportunities	posted_by	1 to 0..*	Each opportunity records the recruiter or Placement Officer who posted it.

No.	Parent Entity	Child Entity	Child FK	Cardinality	Meaning
13	opportunities	opportunity_edit_history	opportunity_id	1 to 0..*	Each edit row references one opportunity.
14	users	opportunity_edit_history	edited_by	1 to 0..*	Each edit row references the user who made the change.

These rows list only explicitly declared foreign keys.

8. Relationship and Cardinality Table (2 of 2)

No.	Parent Entity	Child Entity	Child FK	Cardinality	Meaning
15	opportunities	opportunity_deletion_history	opportunity_id	1 to 0..*	Each deletion row references the opportunity that was deleted or restored.
16	users	opportunity_deletion_history	deleted_by	1 to 0..*	Each deletion row references the user who deleted the opportunity.
17	student_profiles	applications	student_id	1 to 0..*	Each application references one student; a student may submit many applications.

No.	Parent Entity	Child Entity	Child FK	Cardinality	Meaning
18	opportunities	applications	opportunity_id	1 to 0..*	Each application references one opportunity; an opportunity may receive many applications.
19	student_resumes	applications	resume_id	1 to 0..*; child FK optional	Optional reference to the resume attached to the application (nullable).
20	users	applications	applied_by	1 to 0..*; child FK optional	Optional reference to the Placement Officer who applied on behalf of the student (nullable).
21	applications	application_status_history	application_id	1 to 0..*	Each status-history row references one application.
22	users	application_status_history	changed_by	1 to 0..*	Each status-change row references the user who made the change.
23	applications	interviews	application_id	1 to 0..*	Each interview references one application; an application may have several rounds.

No.	Parent Entity	Child Entity	Child FK	Cardinality	Meaning
24	users	interviews	scheduled_by	1 to 0..*	Each interview references the user who scheduled it.
25	applications	offers	application_id	1 to 0..1	Each offer references one application; the FK is UNIQUE, so an application has at most one offer.
26	users	offers	released_by	1 to 0..*	Each offer references the user who released it.
27	users	dashboard_preferences	user_id	1 to 0..*	Each preference row references one user; UNIQUE(user_id, widget_name) prevents repeats.
28	users	report_history	generated_by	1 to 0..*	Each report row references the user who generated it.

8.1 Relationship Notes

The applications table provides the database implementation of a many-to-many association between students and opportunities: each row has one student_id and one opportunity_id, and the pair is declared UNIQUE. The schema uses application_id as its primary key rather than a composite primary key, so interviews, offers and status history can reference a single application directly.

users has a one-to-one relationship with student_profiles and with companies through UNIQUE, NOT NULL foreign keys: a student or recruiter account has exactly one profile record, and a profile cannot exist without an account. The offers table is linked to applications with a UNIQUE foreign key, which limits an application to at most one offer.

9. Database Integrity and Design Observations

Primary keys

Each of the 16 unique tables declares an integer primary key with AUTO_INCREMENT. Every table also keeps created_at and updated_at timestamps, and several history tables store event-specific timestamps such as changed_at, deleted_at and restored_at.

Foreign-key integrity

The SQL declares 28 foreign-key constraints. They ensure that values in the child columns reference existing rows in the parent table. Required foreign keys tie each child row to a parent; the nullable exceptions are user_access_history.action_by, student_profiles.verified_by, student_resumes.removed_by, applications.resume_id and applications.applied_by, which are filled only when the action has taken place.

Keys and uniqueness

users.email, users.phone and users.enrollment_no are UNIQUE (email and phone are also NOT NULL); login_sessions.session_token is UNIQUE and NOT NULL; student_profiles.user_id and student_profiles.enrollment_no are UNIQUE and NOT NULL; companies.user_id is UNIQUE and NOT NULL; offers.application_id is UNIQUE and NOT NULL. The composite constraints UNIQUE(student_id, opportunity_id) and UNIQUE(user_id, widget_name) stop repeated applications and repeated dashboard widgets.

Design decisions and limitations

- users stores role-specific registration fields in one table, so many of those columns are NULL for users of other roles. This keeps registration simple and is recorded without change.
- student_profiles repeats enrollment_no, branch and semester from users so that eligibility checks do not depend on the account table; the values are copied when the profile is created.
- Status and type columns (for example users.status, opportunities.type and applications.status) are VARCHAR columns whose allowed values are enforced by the application rules, not by ENUM or CHECK constraints.
- opportunity_deletion_history stores a copy of the title so that the deletion record stays readable even if the opportunity is later changed.

- Deletion of users and opportunities is a status change (soft delete); no table in the schema uses ON DELETE CASCADE, so child records are never removed automatically.
- Business rules such as “an interview needs a Shortlisted application” and “an offer needs a Selected application” are implemented in the SQL conditions of Industrial Assignment 5, not as database constraints.

10. Conclusion

The ER diagrams document the Student Internship and Placement Management Portal schema through its 16 unique entities and 28 explicitly declared foreign-key relationships. The inventory, overall diagram, module diagrams and cardinality table provide a consistent map of the declared keys, attributes and constraints. The design observations are kept visible so that this assessment can serve as an accurate reference for implementation (Spring Boot entities in Industrial Assignments 10 and 11), validation (Postman testing in Industrial Assignment 12) and future database maintenance.

Primary schema references: Industrial Assignment 5, Database Queries (CREATE TABLE definitions) and Entity and Attribute Dictionary. Project context is drawn from Industrial Assignments 1–4.